

Highlights

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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- Voxel coupling averaged over the source cell equals the continuum's exactly
- The cube's single-site T-matrix follows from distributional moment integrals
- The voxel scheme's leading error is known in closed form for every grid
- A first-moment voxel converges at fourth order for P, SV and SH waves
- In a random stratified layer, one voxel per model layer reaches 1e-6 accuracy

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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Abstract

We analyse the convergence of voxel discretisations of the elastic volume integral equation, in which a medium is divided into cubic cells, each represented by a single-site T -matrix and coupled through the background Green's tensor: the elastic counterpart of the discrete dipole approximation. Because cubes tile space, a layer of identical voxels is exactly a homogeneous layer, so it isolates the discretisation error, free of modelling and shape error. We show that (i) the cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling equals the continuum's and the lattice kernel at normal incidence is in closed form; (ii) the single-site T -matrix follows from distributional moments of the Green's tensor and, against exact scattering by a sphere, is exact statically and departs only by a real, closed-form $(ka)^2$ term; (iii) its self term cancels, so the scheme is a collocation of the continuum equation, of second order with leading error $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission, which no scalar correction can remove; (iv) a Galerkin voxel carrying the first moment of its internal field converges at fourth order, with leading error $-(kd)^4/720$, for incident P, SV and SH

waves at normal and oblique incidence, and its spectral lattice sum needs no Ewald splitting. In a randomly stratified layer the orders are unchanged, and the fourth-order voxel reaches an accuracy of 10^{-6} with one voxel per model layer. Every result is derived symbolically and verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale. The models that describe it are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation represents the sampled model as cells of constant properties and treats every cell as a scatterer: a voxel carrying a single-site T -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green’s tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the

17 single voxel's response is matched to cancel it, refinement converges to the
18 wrong answer without revealing it.

19 *Why a homogeneous layer.* This paper therefore starts from the simplest
20 configuration in which a voxel scheme runs in full (every voxel, every coupling,
21 the whole multiple-scattering series, at normal and at oblique incidence) and
22 the exact answer is nevertheless known: a layer filled with identical voxels.
23 Because cubes tile space, such a layer does not approximate a homogeneous
24 layer; it *is* one. Any difference between its reflection and transmission and
25 those of the continuous layer is therefore error of the discretisation alone, with
26 no modelling error and no shape error behind which to hide. A sphere, the
27 usual test object for volume-integral and discrete-dipole methods [1], cannot
28 give this: its error mixes the discretisation with the staircase approximation
29 of a curved surface by cubes. The layer plays the part of the patch test in
30 finite elements: reproducing a uniform medium correctly is necessary, though
31 not sufficient, for a scheme to be trusted in a heterogeneous one. What the
32 layer establishes about the scheme's two ingredients, the coupling between
33 voxels and the response of a single voxel, are properties of the ingredients,
34 which are the same in any medium: that the averaged coupling is exact, that
35 the self term cancels, and the form of the single-site error. §9 confirms that
36 they carry over to a stratified medium; a laterally heterogeneous one remains
37 to be tested.

38 *What the reader gains.* Four things that a heterogeneous test cannot provide.
39 First, a proof, exact at normal incidence, that the voxel scheme converges
40 to the continuum, and the specific ingredient, a coupling averaged over the
41 source voxel, without which it converges to the wrong answer. Second, the

42 leading error in closed form for every discretisation, so that a grid can be
 43 chosen for a required accuracy before the calculation is run. Third, a single-
 44 site T -matrix derived from first principles, with its error stated and checked
 45 against the exact solution for a sphere. Fourth, a voxel that converges at
 46 fourth order, also in a randomly stratified model (§9): for a reflection ac-
 47 curacy of 10^{-6} it allows voxels 58 times larger in each direction than the
 48 standard one (§10).

49 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*
 50 *space*, so a uniform region discretised into cubes is still that uniform region,
 51 whereas spheres or point scatterers can only approximate it and carry a pack-
 52 ing error of their own. Their *regular lattice* makes the interaction between
 53 cells translation invariant, so the coupling is a convolution (Toeplitz between
 54 planes, Bloch-periodic within them), and the whole system is solved by fast
 55 transforms. And a cube small compared with the wavelength has a *single-site*
 56 *response in closed form*, the elastic depolarisation of a cube (§6), so the only
 57 approximation a voxel introduces is the internal field it assumes.

58 *Multiple scattering on a lattice.* Replacing a heterogeneous body by a lattice
 59 of small scatterers that interact through the background Green's function
 60 is the multiple-scattering description of Foldy [2] and Lax [3]. In electro-
 61 magnetics its discretised form is the discrete dipole approximation [DDA;
 62 4, 5], whose accuracy has been studied in detail: each dipole's polarisability
 63 must account for the local field of the surrounding lattice, which the lattice
 64 dispersion relation fixes so that an infinite lattice propagates waves as the
 65 continuum does to a given order in kd [6]; the error of the method converges
 66 at second order in the dipole spacing when shape errors are absent [1]; and

67 integrating the Green's tensor over each cell rather than sampling it at the
 68 centre improves accuracy, most of all for strong contrasts [7]. The elastic
 69 problem adds a structure the scalar and electromagnetic ones lack. A cubic
 70 voxel responds to the incident field with a force (through its density contrast)
 71 and a moment (through its stiffness contrast), so the state at a voxel has nine
 72 components (displacement and strain), and the single site's self-interaction is
 73 the elastic depolarisation of a cube, the discrete counterpart of the inclusion
 74 tensor of Eshelby [8].

75 *This paper.* The approach originates in the two-and-a-half-dimensional for-
 76 mulation of Nestor [9]: a medium divided into cells that interact through the
 77 background Green's tensor, each carrying a single-site contrast renormalised
 78 by its own near field, with the multiple-scattering series summed. This pa-
 79 per extends it to cubic voxels. §2 sets out the discrete medium and §3 the
 80 continuous layer it must reproduce. §4 shows why the point coupling fails,
 81 and §5 why the source-averaged coupling succeeds exactly. §6 derives the
 82 single-site T -matrix from moment integrals, tests it against the exact sphere,
 83 and shows that with it the discrete system is a collocation of the continuum
 84 equation. §7 gives that collocation's second-order error in closed form, and
 85 §8 the fourth-order voxel, for incident P and S at normal and oblique inci-
 86 dence. §9 repeats the tests on a randomly stratified layer, and §10 discusses
 87 what the results imply for voxel schemes in general. Every result is derived
 88 symbolically in Mathematica and checked against an independent Python
 89 implementation or an exact solution.

2. Setting

The discrete medium. Space is tiled by cubes of side d (half-side $h = d/2$). Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates are ordered (z, x, y) with z down. The field at a voxel is the nine-component state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

displacement followed by engineering strain. A voxel responds with a force and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

with ψ^0 the incident state. Strain rows of $\mathbf{K}$ are receiver derivatives of the point Green's tensor $\mathbf{G}$; moment columns are derivatives with respect to the receiver coordinate, symmetrised with a factor $\frac{1}{2}$ on the shear components. In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta\rho \mathbf{I}_3, \Delta\mathcal{C}), \quad (3)$$

where $\Delta\mathcal{C}$ is the 6×6 contrast stiffness with $\Delta\lambda + 2\Delta\mu$ and $\Delta\lambda$ in the normal block and $2\Delta\mu$ (not $\Delta\mu$) on the shear diagonal.

The laterally uniform layer. A layer of thickness D is filled with n planes of voxels, $d = D/n$, and illuminated at normal incidence. The field is then

109 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 110 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 111 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (4)$$

112 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 113 join the centres of the voxels in one plane (a square lattice of pitch d), and
 114 m is the separation of the two planes in units of d (Fig. 1a,b). The question
 115 is whether, and how fast, this chain converges to the continuous layer as
 116 $n \rightarrow \infty$ at fixed D .

117 *Implementations.* Every result is derived symbolically in Mathematica and
 118 compared with an independent numerical implementation or an exact solu-
 119 tion. “The package” below is the Python implementation of the voxel scheme
 120 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 121 propagator, which serve both as the object under test and, where validated,
 122 as a second implementation.

123 3. The continuous layer and its thin-layer expansion

124 At normal incidence a plane force drives a one-dimensional problem per
 125 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 126 Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission
 127 follow from continuity of displacement and traction at its faces; they agree
 128 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 129 P and S (`ContinuumLimit_Reference.wl`).

130 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 131 ground M_0, ρ_0 , the reflection coefficient expands as

$$R = \frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] + O(D^2), \quad (5)$$

132 exact in the contrast (`ContinuumLimit_ThinLayer.wl`): to first order a thin
 133 layer is an interface across which the traction jumps by the excess inertia and
 134 the displacement by the excess compliance. At oblique incidence the same
 135 statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$ for the P-SV state
 136 $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$,
 137 which reproduces the layer's $O(D)$ reflection coefficients R_{PP}, R_{PS} to 10^{-12} ;
 138 the series truncated at $O(D^k)$ converges at exactly order k ; and a soft thin
 139 layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η tends to the linear-
 140 slip interface as $D \rightarrow 0$. These expansions are the target a single plane of
 141 voxels must reproduce order by order.

142 4. The specular sums of the point Green's tensor

143 *Why transform the sum.* The specular sum (4) adds, at a receiver, the field
 144 of every voxel of a plane m planes away. In that form it is a poor object to
 145 study. The Green's tensor decays slowly with distance, so the sum converges
 146 slowly, and it hides the one question that matters here: which part of it is
 147 the continuum, and which part is the discretisation. The Poisson summation
 148 formula [10] answers both at once, and is the standard tool for lattice sums
 149 of wave fields [11]. For a function f on the plane and the square lattice of

150 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (6)$$

151 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 152 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 2).
 153 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (7)$$

154 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 155 between planes a distance z apart. Each term has a physical meaning. Trans-
 156 formed laterally, the field of a plane of sources is a superposition of plane
 157 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 158 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 159 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 160 far beyond the wavenumber of the field when the voxels are small compared
 161 with the wavelength, so it is evanescent and decays away from the plane as
 162 $e^{-|\mathbf{g}|md} = e^{-2\pi m \sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 163 the lattice sum exactly into the continuum and a discretisation correction,
 164 and the correction converges after a few orders.

165 *The local term that survives.* In the strain-from-moment block the evanescent
 166 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 167 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 168 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 169 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all

170 planes in slab order, the static lattice sum of the point kernel is exactly
 171 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 172 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 173 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 174 reproduces the continuum only if that remainder is cancelled by the voxel's
 175 own self term.

176 5. The tiling identity

177 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$
 178 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 179 Each Poisson order in Eq. (7) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (8)$$

180 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 181 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 182 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ : a partition of unity (Fig. 3).
 183 Two exact statements follow.

184 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (9)$$

185 the continuum's plane-wave term times the vertical form factor. The pack-
 186 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 187 `AveragedSums.wl`).

188 *Within a plane.* The sum over *all* cells of the plane, the voxel’s own included,
 189 is the field at the mid-plane of a uniform plate of thickness d (Fig. 1c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (10)$$

190 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 191 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 192 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 193 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
 194 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 195 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 196 with every cell integral evaluated analytically (below).

197 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 198 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (11)$$

199 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 200 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 201 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 202 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 203 and the quadrature order are raised; Eq. (11) is now the package’s kernel at
 204 normal incidence.

205 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 206 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 207 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 208 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,

209 which never contain the origin, and there the two-dimensional antiderivative
 210 in the face coordinates is continuous, so the face integral is a sum over the
 211 face's corners. The distributional term at $r = 0$ is carried by the divergence
 212 theorem itself. (A triple antiderivative evaluated at the cube's corners is
 213 *not* usable for repeated derivatives: its arctangent terms jump across the
 214 coordinate planes through the origin, and differentiating twice there discards
 215 exactly that term.) The self term so obtained reproduces the classical closed
 216 form of the cube's depolarisation to machine precision.

217 **6. The single site: the cube T -matrix from moment integrals**

218 Every error of a voxel scheme is made at the single site (§10), so the single
 219 site is derived here rather than imported. The classical route imports it. For
 220 an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a
 221 uniform load uniform, with a depolarisation tensor in closed form [8, 12, 13];
 222 its dynamic generalisation is known for spheres and ellipsoids [14, 15]. For
 223 a cube or a general polyhedron the static field of a uniform eigenstrain is
 224 also in closed form [16, 17, 18, 19], but the strain it produces is not uniform,
 225 so there is no exact uniform-strain inclusion to import. In scattering theory
 226 the same object appears as the self-consistent or effective-field closure of
 227 the volume integral equation [20, 21, 22]: a first-order (Born) scatterer [23]
 228 renormalised by the geometric series of its own near field, the step that turns
 229 a contrast into an effective contrast. Nestor [9] introduced the single site
 230 in this form, summing the series of the cell's self integral, with the singular
 231 near field evaluated over an infinitesimal cell. The moments below evaluate
 232 the same self integral over the finite cube, as a distribution. The elastic T -

matrix of Waterman [24] represents a scatterer in spherical multipoles, the natural basis for a sphere; a voxel in a lattice needs one compatible with the lattice, and the moments over the cube provide it. The construction below makes the approximation of a voxel explicit rather than implicit: it derives the cube's T -matrix from the moments of the elastodynamic Green's tensor over the cube, and identifies the uniform-strain closure as one truncation of an exact hierarchy.

6.1. The closed hierarchy

Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q . The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (12)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (13)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (14)$$

all fields evaluated at the centre. Its coefficients are six moments of the background Green's tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \end{aligned} \quad (15)$$

248 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
 249 reaches **N** and **Q** only, and no symmetry, parity or scaling check can detect
 250 it, so it is fixed by convention. The moments are graded by (D, W) , the
 251 number of derivatives and the degree of the weight: **G** $(0, 0)$, **K** $(0, 2)$, **N** $(1, 1)$,
 252 **M** $(2, 0)$, **P** $(2, 2)$, **Q** $(3, 1)$.

253 6.2. The moments as distributions

254 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (16)$$

255 [25, 26]. Expanding g_c in powers of r reduces every moment in Eq. (15) to a
 256 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (17)$$

257 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 258 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 259 gral depends on the shape of the region excluded around the origin. This is
 260 the source of the δ -function term of the inclusion problem, which a numerical
 261 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 262 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
 263 are distributional from the start, and each is moved onto the voxel's faces by
 264 pairing with the test function $w \mathbf{1}_V$, using $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$:

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (18)$$

265 The surface term is evaluated on the faces, a distance h from the origin, where
 266 every derivative of r^m is smooth. The recursion lowers D and W together

and ends on convergent integrals. The face integrals are done in closed form by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (19)$$

which the face route satisfies exactly and a spherical excision returns as 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact. Because every power of r is paired this way, the dynamic moments keep the even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones. A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses those terms, at a real $O((kh)^2)$ cost (§6.4).

6.3. Parity, and the nine-component block

The cube is centrosymmetric. A moment that couples a field quantity of degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (13) contains the first gradient alone: ∂u is determined by its own block and couples to neither u nor $\partial\partial u$. The leading scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which needs only ∂u . The leading far field of the stiffness contrast is therefore fixed by Eq. (13), and *no treatment of the second-gradient block (14) can change it*. What Eq. (14) contributes reaches the far field only through the $O((kh)^2)$ coupling in Eq. (12). That is the order, and the origin (the variation of the field across the voxel), of the collocation error of §7, which the first-moment voxel of §8 removes.

290 With $\Delta\rho = 0$, Eq. (13) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 291 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 292 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are
 293 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (20)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (21)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (22)$$

294 The density channel is the displacement block of Eq. (12) with the second
 295 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 296 of $\mathbf{G}$.

297 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 298 tional channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 299 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 300 latational channel, Eq. (20), is the sphere's exactly: its eigenvalue involves
 301 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 302 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 303 the shape. The cube's shear channels differ from each other. Their isotropic
 304 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (23)$$

305 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 306 order is therefore a pure cubic anisotropy. It splits the deviatoric channel

about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g , and the diagonal shear is the more depolarised. At the background of §7 ($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is 0.4960.

Where Eshelby enters. Equations (20)–(22) follow from the moment integrals and the group theory alone. No ellipsoidal result is used, and no δ term is added by hand. What corresponds to Eshelby is the *truncation*. Keeping ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by the uniformity theorem that ansatz is exact if and only if the inclusion is an ellipsoid. For a sphere the construction is therefore exact in the static limit, and for a cube it is an approximation. The hierarchy (12)–(14) states that approximation as the omitted blocks, rather than as an unquantified shape correction. A polynomial expansion about the centre cannot represent the edge and corner concentrations of an isolated cube's internal field at any finite degree. A voxel in a uniform layer has no such concentrations: its neighbours have the same properties, so its internal field is the continuum's, which is smooth. That is why a low-degree internal field suffices in a lattice (§§7–8) although it is only an approximation for an isolated cube.

Against the exact sphere. These statements are tested against the exact elastic Mie solution for a sphere, with no Eshelby formula in between. The concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilatational, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface route of Eq. (18) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet

332 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
 333 isotropic shear average, Eq. (23), is the depolarisation read from Mie's E_2
 334 to 10^{-8} . The same moment taken as a principal value, over the ball with
 335 a small sphere excluded around the origin, has zero depolarisation in every
 336 channel. The whole static response of the sphere is the δ term.

337 At finite frequency the closure uses the dynamic ball moments, which are
 338 in closed form. It departs from Mie at second order in ka and first order in
 339 the contrast, and the departure is closed-form too: the ball's moments are
 340 elementary, and Mie's low-frequency series with the contrast kept exact gives
 341 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta \lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta \rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta \mu}{\mu}, & n = 2, \end{cases} \quad (24)$$

342 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 343 the radiation damping, agree exactly in all three channels. So the closure's
 344 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 345 own. That is the order the hierarchy assigns to the omitted second gradi-
 346 ents (§6.3), and it has the same origin, the variation of the field across the
 347 scatterer, as the collocation error of §7. Eq. (24) is derived symbolically
 348 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 349 the numerical Mie solve and an independent implementation of the closure
 350 to 5×10^{-4} , which is the size of the next order in the contrast.

351 6.4. The collocation closure

352 The collocation closure takes the internal field of a voxel as uniform, the
 353 truncation of §6.3 for the density and stiffness channels together. The voxel's
 354 own field at its centre is then $\mathbf{S}_{\text{self}}\Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 355 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 356 So $e = \psi + \mathbf{S}_{\text{self}}\Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)^{-1}. \quad (25)$$

357 With the kernel (11) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 358 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (26)$$

359 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 360 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (9), (10)),
 361 so Eq. (26) is precisely the collocation of the continuum Lippmann–Schwinger
 362 equation with a strain piecewise constant over voxels. It does not contain
 363 $\mathbf{S}_{\text{self}}$ at all.

364 The package's cube T -matrix is this closure statically: both blocks agree
 365 with Eq. (25) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 366 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$
 367 through its far-field form-factor corrections. Its self integrals are exact in the
 368 static and radiation parts but omit the real even-power terms of the Green's
 369 tensor's expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 370 self term, whose series agrees with the closed-form elastodynamic tensor to
 371 10^{-14} . By Eq. (26) that error cannot enter the closure scheme.

372 7. Convergence, and the closed-form error

373 *Second order.* For a layer $D = 2$ m with the contrast $\Delta\lambda = 2$ GPa, $\Delta\mu =$
 374 1 GPa, $\Delta\rho = 100$ kg m $^{-3}$ in a background $\alpha = 5000$, $\beta = 3000$ m s $^{-1}$, $\rho =$
 375 2500 kg m $^{-3}$, illuminated by a P plane force, the reflected displacement of
 376 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
 377 16 at $\omega = 60, 300$ and 600 rad s $^{-1}$, with the collocation (26) and with the
 378 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
 379 $k_S h$ alone.

380 *The leading error in closed form.* In Eq. (26) the kernel is integrated exactly;
 381 the only approximation is the internal field's variation across a voxel. At first
 382 order in the contrast the discrete scattered field is therefore a midpoint rule
 383 applied to the continuum's Born integral, with the kernel's cell integral exact.
 384 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 385 it is flat. With k the background wavenumber of the wave concerned (k_P for
 386 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 387 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (27)$$

388 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 389 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 390 and those of Eq. (30) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 391 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
 392 the transmission constant grows to -0.092 .

393 *Why a uniform-field single site stops at second order.* The two constants have
 394 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 395 cancel both. The package *T*-matrix illustrates it: its form-factor corrections
 396 reduce the reflection error by a factor 4.45 and *increase* the transmission
 397 error by 1.7, at every n . The missing term in Eq. (27) is the first moment
 398 of the internal field over each voxel, which radiates through the derivative of
 399 the kernel. A fourth-order scheme must carry it: a single site with a gradient
 400 channel, and a kernel whose cell integrals include the kernel's derivative.

401 8. Fourth order: the first-moment voxel

402 Let each voxel carry the mean *and* the first moment of its internal field.
 403 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_0 = 1$,
 404 $\phi_1 = s/h$, the field is $w = \sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and
 405 strain of the P channel, and the polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with
 406 $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (28)$$

407 with the same two functions on every voxel gives

$$\left(\int_i \phi_a^2 \right) c_{a,i} = \int_i \phi_a w^0 + \sum_{j,b} \left[\int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (29)$$

408 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
 409 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$ and
 410 the local term contributes $-\int_i \phi_a \phi_b / M_P$. Every entry is closed-form. Keeping
 411 only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 converges at fourth

412 order in both reflection and transmission (`ContinuumLimit_FourthOrder.`
 413 `w1`; Table 1). A single plane of first-moment voxels represents the 2 m layer

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (29). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

413
 414 6500 times more accurately than a single plane of uniform ones. The collo-
 415 cation column is the scheme of §6 and reproduces the chain of the full 9×9
 416 system to all printed digits.

417 *The leading error in closed form.* Without scattering, the Galerkin solution
 418 is the L^2 projection Π of the incident field onto each voxel's basis, and the
 419 kernel is integrated over each voxel exactly; at first order in the contrast the
 420 error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field
 421 varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} ,
 422 $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and
 423 every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] \, ds}{\int_{-h}^h e^{i(q+k)s} \, ds}, \quad (30)$$

424 with Π the centre value for collocation, the cell mean for the mean-only voxel
 425 and the projection onto $\{1, s/h\}$ for the first-moment voxel (`ContinuumLimit_`

FourthOrderError.wl). The results are in Table 2. The chain of Eq. (29)

Table 2: The leading error of the discrete layer in closed form, Eq. (30): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$

426

427 reproduces all six forms at $n = 1$ to 8 to 10^{-4} of the error, the remainder
 428 being the second order in the contrast. The first-moment voxel's leading
 429 error is the same in reflection and in transmission: its projection error is or-
 430 thogonal to everything linear across the voxel, so the leading term no longer
 431 depends on the direction in which the field is observed.

432 *In three dimensions.* The three-dimensional first-moment voxel carries the
 433 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns
 434 in all, tested with the same functions and coupled by cell-to-cell double in-
 435 tegrals of the Green's tensor. For a laterally uniform layer at normal in-
 436 cidence it reduces *exactly* to Eq. (29) (ContinuumLimit_Reduction3D.wl).
 437 The lateral first moments are never excited: the lattice and the incident
 438 field are symmetric under reflection about every voxel's centre planes, and
 439 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 440 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-
 441 ishes on the reciprocal lattice and the plane sum of the double integrals
 442 is the plate-to-plate integral in z alone. Two facts mark where genuinely

three-dimensional structure begins. The lateral first moment's form factor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the reciprocal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift, $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h / p\pi + O(k_x^2 h^2)$, so the tiling identity holds only to $O(k_{\parallel} h)$.

Oblique incidence. The scheme is built spectrally. By Poisson summation the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (31)$$

with L_a the lateral Legendre factors. The whole-space tensor decays as $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[i\alpha/2\gamma + i\beta \text{sgn}(z)/2] e^{i\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20° and 40° (`ContinuumLimit_Oblique.wl`; Table 3). The mean-only voxel is second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel} h)$ and the evanescent coupling of the gradient channels do not cost an order. At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (29) to all printed digits.

Incident S. Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (29) with $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes

Table 3: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (31). At 40° the orders are again 2 and 4.

n	mean only		mean and first moments	
	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

465 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
 466 transmission, and the full 9×9 construction with an incident SH wave re-
 467 produces that one-dimensional result to all printed digits. At oblique inci-
 468 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
 are those of Table 4. The orders, and with $k = k_S$ the closed-form constants

Table 4: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

469

470 of Eqs. (27) and (30), are those of the incident P wave.

471 The spectral construction is implemented twice. The second implementa-

tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the definition of the scheme: it takes the depth dependence of the transformed kernel from an independent residue construction, obtains the local weight as a numerical large- k_z limit of the inverted Christoffel matrix, evaluates every double integral by quadrature and separates P from S by sampling the upgoing field at two depths. The raw reflected displacements of the two agree to 9×10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and 20° , $n = 1, 2, 4$, with and without the first moments. Figure 4 collects the convergence.

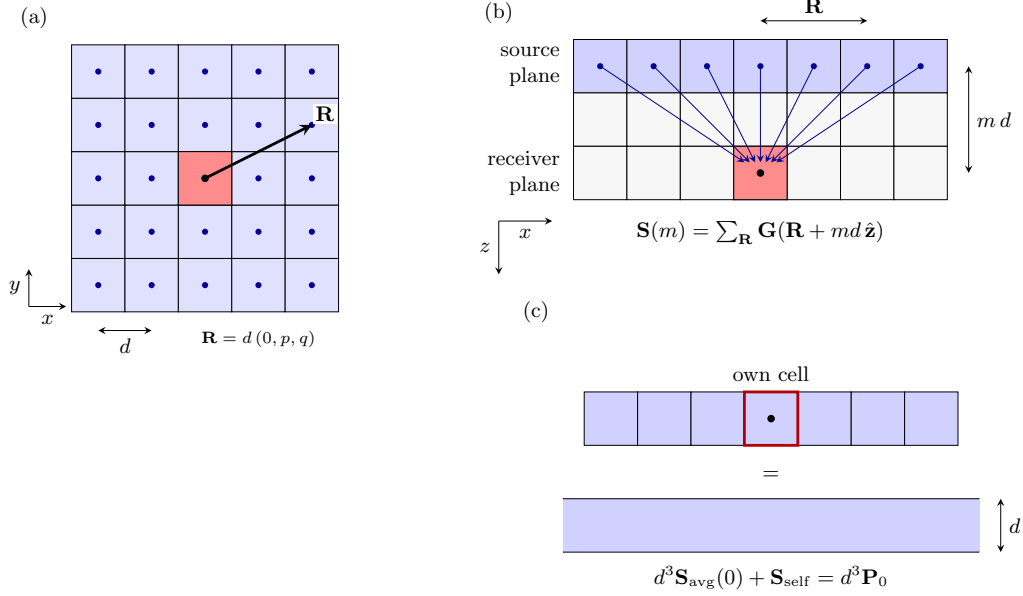


Figure 1: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (red) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (red) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (4). (c) Within a plane: the sum over every cell of the plane, the receiver's own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (10).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation md marked. Below it, a single row of voxels with one outlined as the receiver's own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

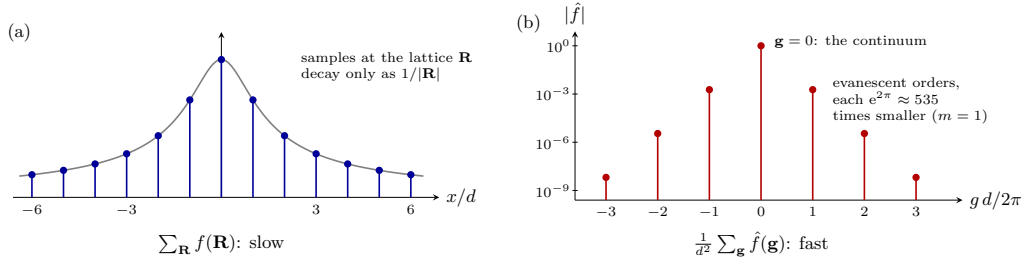


Figure 2: The Poisson summation formula (6) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

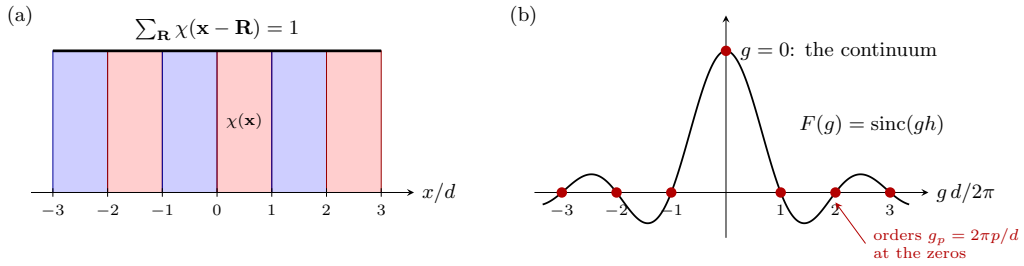


Figure 3: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (7) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

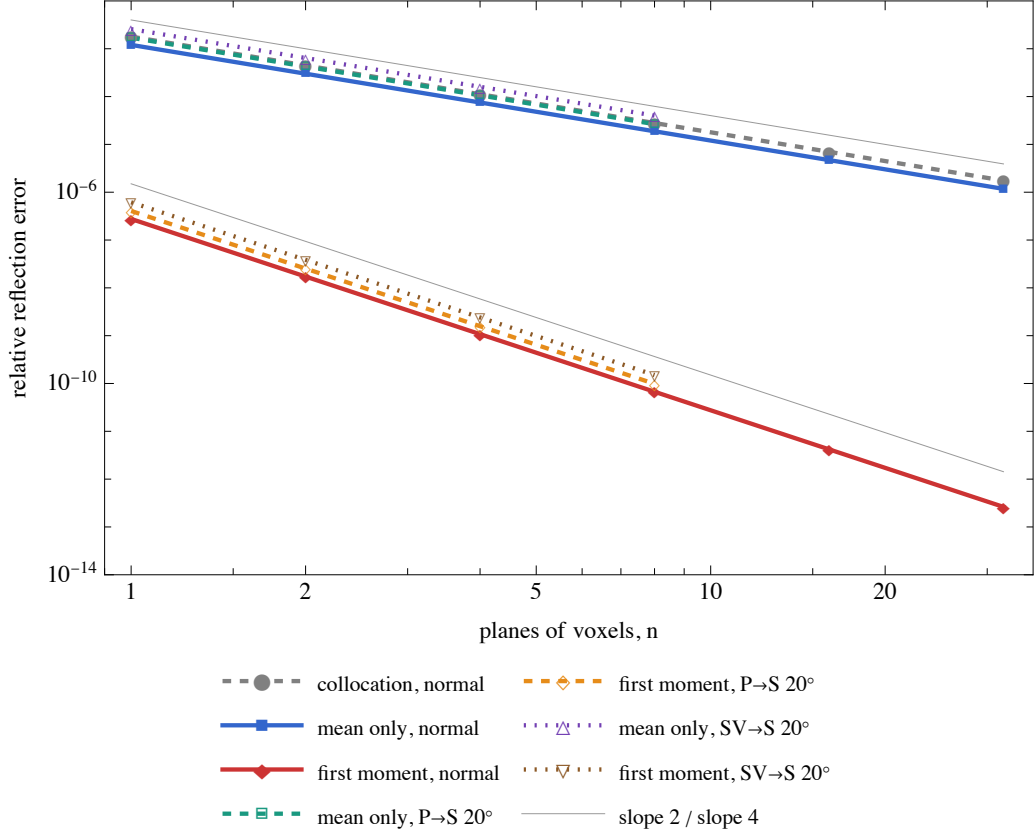


Figure 4: Relative reflection error of the discrete layer against the number of voxel planes n ($D = 2$ m, $\omega = 300$ rad s $^{-1}$, the contrast of §7). Filled: normal incidence, P reflection: collocation (§6), the mean-only voxel and the first-moment voxel of Eq. (29). Dashed, open squares and diamonds: P incident at 20°, the converted S wave, mean-only (green) and first-moment (orange) voxels of Eq. (31). Dotted, open triangles: SV incident at 20°, the reflected S wave, mean-only (purple) and first-moment (brown). Grey: slopes 2 and 4.

Alt text: A log-log plot of relative reflection error (vertical axis, 10^{-14} to 10^{-2}) against the number of voxel planes (horizontal axis, 1 to 32). The collocation and mean-only curves fall as straight lines of slope -2 from about 10^{-3} at one plane; the first-moment curves fall as straight lines of slope -4 from about 10^{-6} at one plane to about 10^{-13} at 32 planes. The oblique P-to-S and incident-SV curves lie alongside the normal-incidence curves of the same scheme, and grey guide lines show slopes 2 and 4.

481 9. A heterogeneous layer

482 A heterogeneous medium is not a layer of identical cells. The simplest
483 heterogeneous case with an exact answer is a stratified model built from layers
484 of constant properties, with real interfaces between them: the form of many
485 layered crustal velocity models. The last test takes such a model. A layer
486 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
487 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
488 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
489 stratification with physical contrasts. The voxels refine the model's layering,
490 with m planes of voxels in each model layer, so no voxel straddles a model
491 interface: the natural discretisation when the model's interfaces are known.
492 At $\omega = 1500$ rad s $^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
493 reverberates between the model interfaces. The exact answer is the product
494 of the model layers' propagators. Its same-type reflection coefficients agree
495 with those of the reflection–transmission recursion of Kennett [27], computed
496 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
497 now changing from plane to plane; nothing else changes. The errors are in
498 Table 5.

499 *The interfaces cost nothing.* Every order is the one measured for the uniform
500 layer. At a model interface the strain of the true field jumps. A voxel basis
501 that is discontinuous from voxel to voxel represents that jump exactly when
502 the interface lies on a voxel boundary, so the interface adds no error of its
503 own. The only requirement is that voxel boundaries coincide with the model's
504 interfaces. A voxel that straddled an interface would carry a contrast that is
505 not uniform within it, a case not examined here.

Table 5: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

506 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
 507 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
 508 layer. The first-moment voxel needs 8, one per model layer: the model's own
 509 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
 510 against 36), that is 3123 against 288 per vertical column of voxels, a factor
 511 of eleven. In three dimensions, with the lateral grid refined as the vertical
 512 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
 513 unknowns by about 2×10^4 . The same model has been solved independently
 514 in Python, with a contrast for each plane: the raw reflected displacements
 515 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
 516 at 20° , with and without the first moments. Lateral heterogeneity needs the
 517 full three-dimensional solver; it is the first extension discussed below.

518 10. Discussion

519 *Where the error of a voxel scheme lives.* The results separate the two ap-
 520 proximations a voxel scheme makes. The *coupling* between voxels makes
 521 none: averaged over the source cell it is the continuum’s cell integral ex-
 522 actly, between planes by the form factor’s zeros and within a plane by the
 523 tiling identity, and the self term it omits is exactly the one the single site
 524 carries. Every error therefore comes from the *single site*, from the internal
 525 field it assumes. For a uniform field that error is the midpoint rule’s, known
 526 in closed form; for a linear field it is the next term of the same expansion.
 527 The converse is the practical warning. A kernel that is not the cell integral
 528 (the point Green’s tensor, or a cell average that does not match the single
 529 site’s closure) leaves a local term that does not vanish as the voxels shrink
 530 (§4): the evanescent lattice orders carry a static contribution of fixed size
 531 per voxel volume. A scheme built that way converges to the wrong limit,
 532 and refinement cannot reveal it, because the error is scale-invariant.

533 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 534 single site from a modelling choice into a derived object with a stated error.
 535 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
 536 only by the cubic split of Eq. (23). Its dynamic part departs from exact
 537 scattering by a sphere only through the real $(ka)^2$ term of Eq. (24); the
 538 radiation damping is exact. That term and the collocation error of Eq. (27)
 539 have the same origin, the variation of the field across the scatterer, and the
 540 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 541 field a voxel must represent is the smooth continuum field, not the field of
 542 an isolated cube, with its concentrations at the edges and corners.

543 *Relation to the discrete dipole approximation.* The same partition appears in
 544 the electromagnetic literature as the choice of polarisability. The Clausius–
 545 Mossotti relation, the lattice dispersion relation [6] and the integrated Green
 546 tensor [7] all adjust the single site so that the lattice reproduces the con-
 547 tinuum. For space-filling cubes the adjustment is neither needed nor a free
 548 parameter: the tiling identity fixes the local field exactly, the self term of the
 549 collocation closure is the one that makes it cancel, and what is left is a closed-
 550 form error rather than an empirical one. The second-order convergence re-
 551 ported for the DDA by Yurkin et al. [1] corresponds to the uniform-field
 552 voxel’s, and the opposite-signed constants of Eq. (27) explain why a polaris-
 553 ability correction can improve one observable only at the expense of another,
 554 as the package’s own form-factor corrections do in reflection and transmis-
 555 sion. Going beyond second order requires a richer internal field, which is the
 556 point of higher-order bases for volume integral equations; the first-moment
 557 voxel is the lowest such basis, and in its spectral form the Galerkin double
 558 average makes the lattice sum converge without Ewald splitting.

559 *The cost of fourth order.* A first-moment voxel carries four times as many
 560 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 561 forms show what that buys. To reach an error ε in reflection, the uniform
 562 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 563 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 564 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 565 number of unknowns by about 5×10^4 (these are the Born-order constants;
 566 at the contrast studied the first-moment voxel’s measured errors lie within
 567 5% of them, while the uniform voxel’s transmission constant grows with the

contrast, so the comparison is if anything conservative). The stratified model of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-moment voxels for the same accuracy. The gain is largest precisely where it matters: when the model varies on scales that are coarse compared with the accuracy the wavefield must have.

Limitations and outlook. The results concern laterally uniform media, a homogeneous layer and a randomly stratified one, for which the discrete system is exactly a chain of planes; a laterally heterogeneous medium needs the first-moment voxel in a full three-dimensional solver, where the gradient channels couple through the evanescent lattice orders (§8) and the tiling identity holds only to $O(k_{\parallel}h)$. The numerical evidence is at moderate, physically typical contrasts (up to 15% in density and moduli) and at $k_{\parallel}h \leq 0.3$, the range in which the single-site T -matrix is validated; the closed forms are first order in the contrast, and the departures from them at the full contrast are of second order. What is established is the *order* of convergence and its leading constant, not the behaviour at strong contrast, where the single site's own closure may limit the accuracy before the lattice coupling does. The single site's dynamic error is in closed form for a sphere only; for a cube, for which no exact solution exists, it is known through the layer (§§7–8). Two extensions follow directly. The first is the three-dimensional solver itself. The second is a stratified *background*, a layered reference medium, as distinct from the stratified contrast of §9. There the lattice coupling then carries the reverberation of the layering, which is not translation invariant in depth, and whether a tiling identity holds for it is the natural next question.

592 11. Conclusions

593 A lattice of cubic voxels reproduces a continuous elastic medium in a pre-
594 cise sense. Because cubes tile the plane, the source-cell-averaged coupling be-
595 tween voxels is the continuum’s cell integral exactly, and at normal incidence
596 the lattice kernel is available in closed form with no lattice sum. The single
597 site is derived rather than assumed. The moments of the Green’s tensor over
598 the cube, taken as distributions, give a closed hierarchy whose first-gradient
599 truncation is the uniform-strain closure. That closure is exact for a sphere
600 in the static limit, and it departs from exact scattering by a sphere only by a
601 real $(ka)^2$ term, with the radiation damping exact. For a cube it differs from
602 the sphere only by a cubic anisotropy of the shear response. The collocation
603 closure carries the same self term as the lattice kernel, which cancels, so the
604 discrete system is a collocation of the continuum equation. It converges at
605 second order, with a leading error that is a midpoint-rule error of the internal
606 field, known in closed form for every discretisation, and opposite in sign in
607 reflection and transmission. A voxel that also carries the first moment of
608 its internal field converges at fourth order, at normal incidence (where the
609 three-dimensional scheme reduces exactly to one dimension) and at oblique
610 incidence, with a leading error, $-(kd)^4/720$, that no longer depends on the
611 direction of observation. The orders survive the interfaces of a randomly
612 stratified model, provided voxel boundaries coincide with its interfaces, and
613 there the first-moment voxel reaches an accuracy of 10^{-6} with one voxel per
614 model layer, where the uniform-field voxel needs about forty. The results,
615 and the evidence for each, are collected in Table 6.

Table 6: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (10)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (11)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (18)	sum rule (19) exact; M, Q equal an independent archive, 81/81
Channels (20)–(22)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (23)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube’s bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (24)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (26)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (27)	$n = 1\text{--}16$, to 10^{-4} of the error
First-moment voxel, Eq. (29)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (31)	order 4, P and S, 20° and 40°
Incident SV and SH, all schemes	orders 2 and 4; constants with k_S to 4×10^{-4}

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618 the Research School of Earth Sciences of the Australian National University,
619 Canberra [9].

620 **Declaration of competing interest**

621 The author declares no known competing financial interests or personal
622 relationships that could have appeared to influence the work reported in this
623 paper.

624 **Data availability**

625 The Mathematica notebooks and Python scripts that derive and check ev-
626 ery result, and the package they test, are archived at [https://doi.org/10.](https://doi.org/10.5281/zenodo.23003241)
627 [5281/zenodo.23003241](https://doi.org/10.5281/zenodo.23003241) and developed at [https://github.com/tmnestor/](https://github.com/tmnestor/MultipleScatteringCalculations)
628 [MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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